

Math 1232 Summer 2025

Mastery Quiz 6

Due Tuesday, July 22

This mastery quiz has three topics. Everyone should do S5, but if you already have a 4/4 on M2 or a 2/2 on S4, feel free to skip these topics. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write “yes” or “no” or give a single number.

Please turn in this quiz in class on Tuesday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 2: Integration Techniques
- Secondary Topic 4: Arc Length and Surface Area
- Secondary Topic 5: Differential Equations

Name: Solutions

Major Topic 2: Integration Techniques

(a) Compute $\int x \arctan(x^2) dx$.

Solution.

$$\begin{aligned}\int x \arctan(x^2) dx &= \frac{x^2}{2} \arctan(x^2) - \int \frac{x^2}{2} \frac{2x}{1+x^4} dx \\ &= \frac{x^2}{2} \arctan(x^2) - \int \frac{x^3}{1+x^4} dx \\ &= \frac{x^2 \arctan(x^2)}{2} - \frac{1}{4} \ln |1+x^4| + C.\end{aligned}$$

(b) Compute $\int \frac{x^3 - x + 2}{x + 1} dx$

Solution. Polynomial long division gives that

$$\begin{aligned}x^3 - x + 2 &= x^2(x + 1) - x(x + 1) + 2 \\ \frac{x^3 - x + 2}{x + 1} &= x^2 - x + \frac{2}{x + 1} \\ \int \frac{x^3 - x + 2}{x + 1} dx &= \int x^2 - x + \frac{2}{x + 1} dx \\ &= \frac{x^3}{3} - \frac{x^2}{2} + 2 \ln |x + 1| + C.\end{aligned}$$

(c) Compute $\int \sin^6(x) \cos^5(x) dx$.

Solution. This is an odd power of cosine, so we set up a $u = \sin(x)$ substitution.

$$\begin{aligned}\int \sin^6(x) \cos^5(x) dx &= \int \sin^6(x) (1 - \sin^2(x))^2 \cos(x) dx \\ &= \int u^6 (1 - u^2)^2 du = \int u^6 - 2u^8 + u^{10} du \\ &= \frac{1}{7} \sin^7(x) - \frac{2}{9} \sin^9(x) + \frac{1}{11} \sin^{11}(x) + C.\end{aligned}$$

Secondary Topic 4: Arc Length and Surface Area

(a) Compute the arc length of the curve $y = \frac{x^4}{8} + \frac{1}{4x^2}$ as x varies from 1 to 2.

Solution. We have $y' = \frac{x^3}{2} + \frac{-1}{2x^3}$, and thus

$$\begin{aligned} L &= \int_1^2 \sqrt{1 + (x^3/2 - x^{-3}/2)^2} dx = \int_1^2 \sqrt{1 + x^6/4 - 1/2 + x^{-6}/4} dx \\ &= \int_1^2 \sqrt{x^6/4 + 1/2 + x^{-6}/4} dx = \int_1^2 \sqrt{(x^3/2 + x^{-3}/2)^2} dx \\ &= \int_1^2 x^3/2 + x^{-3}/2 dx = \frac{x^4}{8} - \frac{1}{4x^2} \Big|_1^2 \\ &= \frac{16}{8} - \frac{1}{16} - \left(\frac{1}{8} - \frac{1}{4} \right) = \frac{33}{16}. \end{aligned}$$

(b) Set up (but don't compute!) the area of the surface obtained by taking the curve $x = \ln y^2 + 5y$ as y goes from 2 to 7 and rotating it around the y -axis.

Solution. We have $x' = \frac{2y}{y^2} + 5 = \frac{2}{y} + 5$ so

$$L = \int_2^7 2\pi \ln(y^2 + 5y) \sqrt{1 + \left(\frac{2}{y} + 5 \right)^2} dy.$$

Secondary Topic 5: Differential Equations

(a) Find a general solution to the equation $y' = xy e^x$.

Solution.

$$\begin{aligned}\frac{dy}{dx} &= xy e^x \\ \int \frac{dy}{y} &= \int x e^x dx \\ \ln |y| &= x e^x - \int e^x \\ \ln |y| &= x e^x - e^x + C \\ y &= C e^{x e^x - e^x}.\end{aligned}$$

(b) Find the (particular) solution to the initial value problem $\frac{y'}{x} = \cos^2(y)$ if $y(0) = \pi/4$.

Solution.

$$\begin{aligned}\frac{dy}{dx} &= x \cos^2(y) \\ \int \sec^2(y) dy &= \int x dx \\ \tan(y) &= x^2/2 + C \\ y &= \arctan(x^2/2 + C).\end{aligned}$$

Then we have

$$\begin{aligned}\pi/4 &= \arctan(C) \\ \tan(\pi/4) &= C \\ C &= 1 \\ y &= \arctan(x^2/2 + 1).\end{aligned}$$